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Reusability report: multi-sample omics network inference operates in distinct robustness regimes defined by prior knowledge and data availability

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Knowledge-constrained network inference enables the reconstruction of regulatory programmes from high-dimensional biological data, yet the conditions under which shared and context-specific structures can be reliably identified remain poorly understood. Pablo Rodriguez-Mier et al. recently introduced CORNETO, a framework for multi-sample constrained inference. Here we reproduce, evaluate and extend CORNETO in terms of robustness, identifiability and bias–variance trade-offs. Using dynamic haematopoiesis and stable cortical datasets, we systematically assess four network reconstruction paradigms while preserving the original method-specific evaluation procedures and benchmark definitions. We show that multi-sample inference generally improves robustness and reproducibility, although gains in predictive performance depend on biological context and the completeness of the prior network. To enhance stability, we introduce bootstrap aggregation and selective re-expansion strategies that improve consensus robustness and recovery while preserving context-specific signals without modifying the underlying optimization scheme or sacrificing generality. Across datasets and methods, our results reveal distinct inference regimes governed by sample availability, prior knowledge and regularization. Together, these findings position network reconstruction as a regime-dependent learning problem and provide practical principles for robust multi-sample biological network inference.

Inferring regulatory networks from high-dimensional biological data is fundamentally underdetermined, as several candidate regulatory structures may be consistent with the same observed measurements. As a result, predictive performance alone cannot establish that an inferred network corresponds to the underlying molecular regulatory mechanism. In practice, understanding biological systems requires methods that capture context-dependent interactions among genes, proteins and metabolites[1], in which nodes are molecular entities and edges represent functional interactions[2]. Because interactions vary across cell types and conditions, accurate network inference is essential for understanding cellular processes, diseases and treatment responses[3,4]. Yet, several network configurations can explain the same observations, making it ill-posed to distinguish shared and context-specific structure.

To meet the associated challenges, methods for inferring networks fall into two categories: data-driven and knowledge-driven. Data-driven models offer strong predictive power but typically require

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large datasets and are often difficult to interpret[5–10]. Knowledge-driven methods leverage prior-knowledge networks (PKNs), which improves interpretability and enables inference in sparse-data settings, but they are often tailored to specific network types and applied per sample[11–13]. Notably, per-sample inference may complicate the identification of robust shared versus context-specific structure, as several network solutions can explain the same observations under noisy or weakly constrained settings. Although shared patterns can be identified by overlapping independently inferred sample-specific networks, such approaches may remain sensitive to instability and variability. Although the original CORNETO study extensively analysed the advantages and trade-offs of multi-sample coupling, the robustness, extensibility and optimization behaviour of these approaches across distinct biological regimes remain incompletely characterized.

Constrained optimization for the recovery of networks from omics (CORNETO)[14] provides a unified, knowledge-driven framework that reformulates multi-sample network inference as a mixed-integer constrained optimization problem based on network flows. This enables joint inference of shared and condition-specific subnetworks via structured sparsity penalties, thereby stabilizing reconstruction and reducing solution variability and yielding sparser, more interpretable models. Moreover, CORNETO unifies several methods—flux balance analysis (FBA)[15], sparse FBA (sFBA)[16,17], integrative metabolic analysis tool (iMAT)[18], causal reasoning for network identification using integer value programming (CARNIVAL)[19] and prize-collecting Steiner trees (PCSTs)[20]—as special cases of a single formulation, and it provides an open-source implementation that supports modular extensions.

Here we build upon the work of Pablo Rodriguez-Mier et al. by systematically evaluating the computational reusability, robustness, reproducibility and practical behaviour of the CORNETO framework (Fig. 1a,b). Our primary objective is not to systematically validate the biological correctness of the inferred regulatory mechanisms, for which comprehensive experimental ground truths are unavailable, but to characterize when and why multi-sample constrained inference produces stable and reproducible solutions across different inference and data settings. We then extend this evaluation by systematically applying the compatible CORNETO formulations to complementary biological settings: a dynamic haematopoietic differentiation dataset[21] and a comparatively stable adult brain cortex dataset[22]. These systems represent contrasting regimes of transcriptional variability and intersample heterogeneity, allowing us to investigate how biological context influences the behaviour of constrained multi-sample inference; in this selection, the latter case is a stringent test due to the limited gene-expression variability[23,24].

For each case study, we systematically evaluated the computational robustness, stability and sensitivity of CORNETO formulations using iMAT, CARNIVAL, Steiner tree and knowledge-primed neural networks (KPNNs)[25]. Biological relevance was assessed separately and only when established biological relations provided an interpretable reference. Each tool addresses distinct biological questions: metabolic flux consistency, signalling propagation, protein interaction modules and neural-network-based regulatory inference (Fig. 1b). Importantly, these frameworks do not produce directly equivalent outputs and were, therefore, not compared using a single unified ground-truth objective. Instead, each method was evaluated within shared biological contexts using framework-appropriate inputs, optimization objectives and evaluation criteria. Our goal was not to rank methods against a common target but to characterize their robustness, behaviour and sensitivity under controlled multi-sample settings. Inputs, outputs, optimization objectives, prior-network definitions and evaluation metrics for each framework are summarized in Supplementary Table 1. Additionally, to assess performance across different levels of data coverage and network complexity, we tested framework robustness under challenging input conditions (Fig. 1c), including imbalanced subpopulations[26,27], smaller datasets[28] and variations in PKNs[29–31].

Finally, we investigated practical strategies for improving the robustness and interpretability of CORNETO-derived solutions (Fig. 1d). Beyond evaluating the effects of prior-network completeness and sample availability, we introduced a bootstrap aggregation procedure[32] to stabilize stochastic optimization and selective re-expansion to recover informative features lost during aggressive pruning. These adaptations improved recall, interpretability[33] and predictive accuracy while preserving sparsity, demonstrating the flexibility of CORNETO in incorporating modular improvements.

Together, these analyses characterize constrained multi-sample network reconstruction as a regime-dependent learning problem, demonstrating that the benefits of joint inference depend on the interplay among prior knowledge, biological heterogeneity, regularization and data availability, and they provide practical guidance for applying and extending CORNETO across diverse biological contexts.

Results

Evaluating the reproducibility of the reported results

We reproduced the original CORNETO results (Methods, Supplementary Fig. 1 and Supplementary Table 2), confirming that CORNETO accurately reconstructs context-specific subnetworks within prior-constrained optimization spaces derived from multi-sample omics data. As CORNETO is not a single method but rather a framework integrating several optimization- and network-based approaches, we refer readers to Methods and Supplementary Table 1 and to the original article for a detailed description of each component method and, equally important, the associated evaluations used during the initial reproduction analyses. Quantitative evaluations followed the same benchmarking procedures, validation criteria and reference definitions implemented in the original method-specific codebases and benchmark scripts. In all analyses, the PKN defined the constrained, feasible optimization space, consistent with the original implementations, but was not treated as a universal biological ground truth. In the following sections, we explicitly indicate any modifications introduced relative to the original implementation, including changes to input data, prior networks, optimization settings or evaluation metrics; those modifications are also summarized in Supplementary Table 1.

Regarding reproducibility, certain CORNETO tools, including FBA and KPNN[34], are inherently stochastic due to optimization heuristics or neural network training[35]. To assess the impact of this stochasticity on inference stability, we repeated both single- and multi-sample analyses several times using the original datasets and compared outcomes across independent runs.

For FBA, multi-sample sFBA consistently generated smaller networks than single-condition FBA, indicating more compact and robust network inference (Supplementary Fig. 2a). Network reduction depended on the minimum biomass constraint: lower biomass thresholds allow greater flexibility to remove non-essential reactions, whereas near-optimal growth requires other reactions to sustain a high flux. At 10% of the optimal biomass, the networks were reduced by ~32%, whereas at 90%, the reduction decreased to ~6% (Supplementary Fig. 2b). This was accompanied by increased variability across runs, with the coefficient of variation rising from 2.9% to 8.9% (Supplementary Fig. 2c), reflecting greater solution degeneracy near-optimal growth, as several alternative flux distributions can satisfy the constraint.

For KPNN, using the T-cell-receptor stimulation dataset, the scores for the area under the curve of the receiver operating characteristic (ROC-AUC)[36] remained highly stable across repeated runs, both before and after pruning, indicating the minimal impact of training stochasticity on predictive performance (Supplementary Fig. 2d). The per-run performance deviation (ΔROC-AUC < 0.001) was negligible (Supplementary Fig. 2e), further demonstrating the reproducibility of network inference using KPNN under fixed data splits.

In summary, although some CORNETO components involve stochastic optimization or neural network training and are, therefore,

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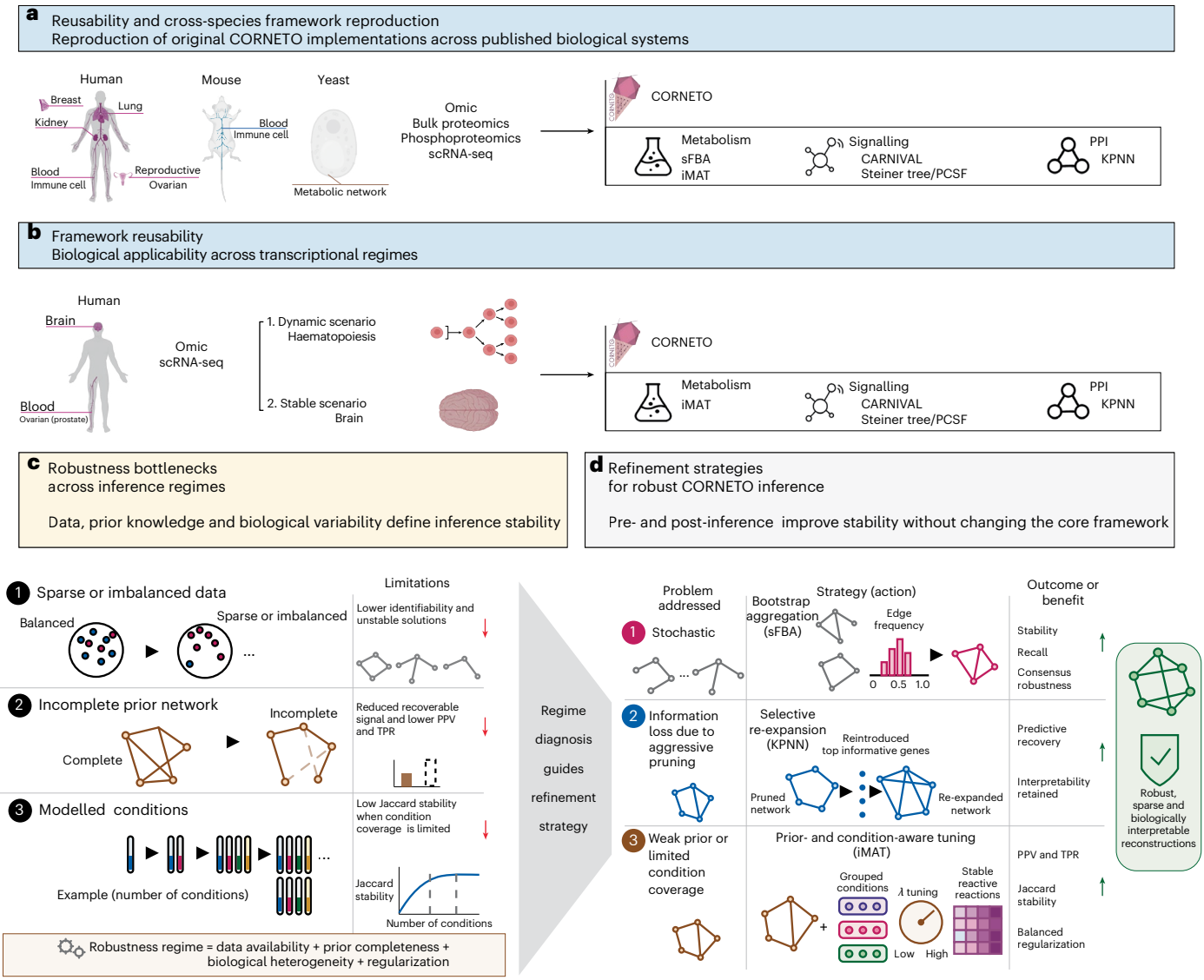


Fig. 1 | Overview of CORNETO framework reproducibility, reusability, constraints and refinement strategies. a, Reproducibility of the CORNETO workflow across multi-omics datasets (proteomics, phosphoproteomics and scRNA-seq) and model organisms, showing integration of metabolic (sFBA and iMAT), signalling (CARNIVAL and Steiner tree/prize-collecting Steiner forest (PCSF)) and PPI-based (KPNN) inference tools. **b**, Reusability of CORNETO in human single-cell datasets spanning dynamic (haematopoiesis) and stable (brain) systems, illustrating four applications: metabolic inference, PPI module discovery, signalling optimization and causal signalling reconstruction. **c**, Robustness bottlenecks across inference regimes. CORNETO performance is influenced by sparse or imbalanced data, incomplete PKNs, and the number of jointly modelled conditions. These factors affect identifiability, the recoverable signal, PPV and TPR, and the Jaccard stability, defining the regime in which constrained network inference operates. **d**, Matched refinement strategies for robust CORNETO inference. The regime diagnosis from **c** guides pre- and post-inference actions, including bootstrap aggregation to reduce stochastic variability and improve consensus robustness, selective re-expansion to recover informative features lost after aggressive pruning, and prior- and condition-aware tuning to balance regularization and improve stable active-network recovery. Together, these strategies improve robustness, sparsity and network interpretability without changing the core CORNETO framework.

sensitive to the initialization seeds, the multi-sample formulation consistently reduced run-to-run variability and produced more stable and reproducible network reconstructions than the corresponding single-sample implementations. These results support the use of multi-sample constrained inference as a robustness-enhancing strategy for integrative, multi-condition omics studies.

Assessing CORNETO across dynamic and stable biological systems

We next evaluated CORNETO as an integrated framework by applying all feasible tools to two biologically complementary contexts: a dynamic system (haematopoiesis) and a steady-state scenario (brain tissue).

The first context represents a biological regime characterized by continuous state transitions. We analysed four single-cell RNA-sequencing (scRNA-seq) samples derived from human CD34$^+$ haematopoietic progenitor cells, annotated into erythroid, monocytic, lymphoid and megakaryocytic differentiation states (Fig. 2a and Supplementary Fig. 3a). We defined cell-type-specific contrasts by treating each cell type as the target condition relative to the others. This shared contrast design was applied uniformly to iMAT, KPNN, Steiner tree and CARNIVAL to evaluate whether the multi-sample approach of CORNETO improves interpretability and robustness in a dynamic developmental context (see Supplementary Table 1 for the overall details).

For iMAT, which infers active metabolic states from gene-expression-derived constraints, we evaluated the regularization weight λ, a key

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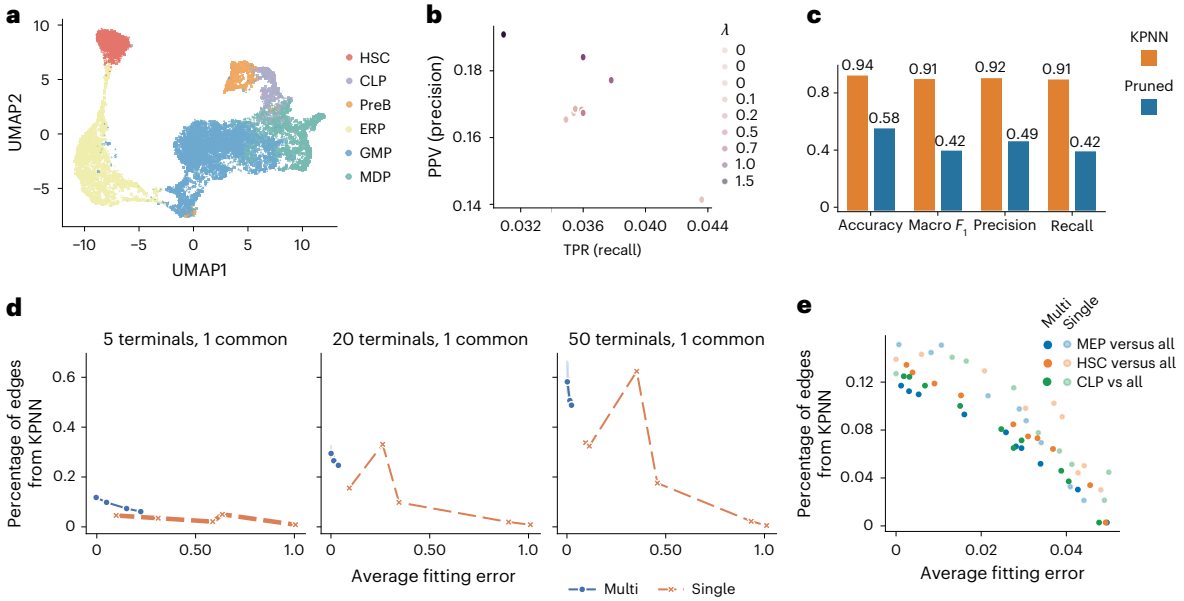


Fig. 2 | Evaluation of CORNETO tools in the haematopoiesis differentiation system. a, Uniform manifold approximation and projection (UMAP) of haematopoietic populations across four donors for HSC, CLP, PreB, GMP, MDP, ERP and MEP. **b**, iMAT precision and recall for several λ values. **c**, KPNN classification accuracy, precision and recall before and after pruning, alongside changes in network size. **d**, Steiner tree reconstructions using 5, 20 or 50 terminal genes, showing fitting error versus recovery of prior-knowledge edges for single- and multi-sample settings. **e**, CARNIVAL reconstructions across all pairwise contrasts, showing fitting error versus recovered edges for several regularization strengths.

parameter governing the multi-sample formulation and its added value relative to independent sample optimization. Specifically, λ controls the extent to which flux distributions are encouraged to be similar across samples (for a low λ, each sample is fitted independently, whereas for a high λ, flux profiles are forced to converge) (Fig. 2b), using the same benchmark definitions and evaluation procedures implemented in the original reference framework ('Evaluation metrics and reference definitions' in Methods). Across λ values, the positive predictive value (PPV) and true positive rate (TPR) remained low with limited variation (maximum PPV ≈ 0.20 and maximum TPR ≈ 0.042). Modest regularization ($\lambda \approx 0.1$–0.25) yielded the best performance, with a slight increase in PPV (~0.17–0.18) and stable TPR (~0.034–0.038). Stronger regularization ($\lambda \geq 0.75$) markedly reduced both metrics, indicating over-regularization. The apparent PPV increase at $\lambda = 1.5$ reflects excessive sparsity rather than improved recovery. These trends and further analyses (Supplementary Fig. 3d,e) support the use of moderate λ values to balance robustness and biological variability.

Conversely, the KPNN, which uses the PKN as a scaffold to construct a cell-type predictive model, achieved strong supervised performance for cell-type classification (accuracy, macro F_1, precision and recall all >0.9; Fig. 2c) using curated cell-type annotations provided with the dataset as the supervised target. Because KPNN learns a shared representation across all cells, it provides an opportunity to assess whether network-constrained compression can retain predictive information derived from several samples while improving model interpretability. Although predictive accuracy and network sparsity do not fully capture the biological objectives of KPNN models, we analysed these quantities as practical indicators of optimization behaviour, representation efficiency and the trade-offs introduced by biologically constrained pruning. After CORNETO-based structural pruning, supervised classification performance decreased substantially (accuracy ≈ 0.58, macro $F_1 \approx 0.42$, precision ≈ 0.49 and recall ≈ 0.42), reflecting the impact of aggressive sparsification. This pruning substantially reduced network complexity, from 9,791 inputs and ~83,000 edges to 66 inputs and 267 edges (Supplementary Table 2). Importantly, pruning also altered feature prioritization: differentiation-associated genes lost prominence, whereas more general metabolic or structural genes gained prominence (Supplementary Fig. 3f,g). These findings suggest that although multi-sample learning captures robust shared signals, aggressive pruning improves interpretability at the expense of cell-type-specific regulatory programmes.

For the Steiner tree task, which identifies a minimum-cost regulatory subnetwork connecting predefined terminal genes via informative intermediate nodes and edges, we evaluated the multi-sample formulation of CORNETO by varying the number of terminals (5, 20 and 50) and comparing multi- versus single-sample inference. Across all settings, multi-sample inference consistently achieved a lower fitting error for the same fraction of active edges within the constrained prior network (Fig. 2d, combined view). This advantage increased with network complexity (20–50 terminals), as multi-sample reconstructions recovered a larger fraction of reference-consistent edges within the constrained prior network (~0.3–0.6) at comparable or lower error (~0.05–0.15). By contrast, single-sample performance degraded sharply as the number of terminals increased (Supplementary Fig. 3b), reflecting sensitivity to noise. Moreover, multi-sample optimization showed smoother convergence and reduced variability (Supplementary Fig. 3c). Joint inference across haematopoiesis replicates yielded up to ×2–3 higher edge recovery without increased error, indicating that shared structural constraints stabilize reconstruction under matched conditions.

We next applied CARNIVAL to infer upstream signalling pathways from differential expression across haematopoietic lineages using the all-versus-all contrast design. To assess the added value of jointly modelling several samples, we compared single-sample and multi-sample formulations across a range of regularization strengths. Multi-sample CARNIVAL consistently achieved a lower fitting error for the same proportion of recovered active edges within the constrained signalling prior (Fig. 2e and Supplementary Fig. 4a). This advantage was most apparent at moderate regularization ($\lambda \approx 0.05$–0.2), as multi-sample solutions recovered substantially more reference-consistent edges within the constrained signalling prior at comparable error. For instance, at $\lambda = 0.1$, the multi-sample model retained ~720 shared edges, compared with ~470–500 in single-sample runs, with similar fitting errors (Supplementary Fig. 4a). As λ increased, both formulations

exhibited the expected sparsity versus accuracy trade-off, but the multi-sample performance curve declined more gradually and remained more stable across regularization strengths, indicating that information sharing across samples improves robustness and preserves signalling structure under stronger regularization.

We then evaluated a second case study based on adult brain cortex single-cell data (Supplementary Note 1), representing a comparatively stable biological system with limited transcriptional variability. In this setting, the biological impact of multi-sample regularization was less pronounced. (1) For iMAT, both PPV and TPR remained low across λ values (maximum PPV ≈ 0.12 and maximum TPR ≈ 0.027), with modest gains at intermediate regularization ($\lambda \approx 0.1$–0.25), reflecting the constrained metabolic dynamic range of steady-state neuronal populations. (2) KPNN achieved strong baseline classification performance (macro $F_1 \approx 0.94$), but CORNETO-based pruning reduced class-balanced metrics (macro $F_1 \approx 0.80$) and shifted feature importance from neuron-specific regulators of neurogenesis and synaptic function (for example, *FOXG1*, *TBR1* and *SATB2*) towards more general transcriptional control genes. (3) Multi-sample Steiner tree inference yielded more stable subnetworks than independently inferred single-sample runs, yielding more coherent subnetworks that emphasized common neuronal regulatory backbones, although without the clear biologically ordered gradients observed in haematopoiesis. (4) CARNIVAL did not return feasible solutions, probably due to extreme intercell-type homogeneity and class imbalance, which generated a highly degenerate optimization space incompatible with its bounded formulation.

Together, these findings suggest that the biological value of the multi-sample framework in CORNETO depends on the structure of the variability present in the data. In haematopoiesis, where differentiation generates coordinated but distinct regulatory states, joint inference reveals developmental relations that are consistently recovered across complementary methods. By contrast, adult brain populations share more conserved transcriptional programmes, limiting the extent to which further biological structure can be extracted through sample coupling. These two case studies illustrate contrasting behaviours of constrained multi-sample inference. For the haematopoietic dataset, characterized by coordinated differentiation-associated variation, sample coupling provided clearer gains in network stability and biological interpretability, whereas for the adult cortical dataset, these gains were more limited. Because each setting is represented by a single dataset, these observations should not be interpreted as establishing general differences between dynamic and homeostatic biological systems. Rather, they motivate the hypothesis that the value of multi-sample coupling depends on the structure and magnitude of biological variability present in the data.

However, because comprehensive ground-truth regulatory interactions are unavailable for these developmental systems, these observations should be interpreted as supportive and hypothesis-generating rather than as definitive evidence of a causal regulatory structure.

Assessing the biological value of multi-sample network inference across diverse cellular contexts

To determine whether the inferred networks captured biologically meaningful organization rather than merely improving the optimization metrics, we examined how the resulting subnetworks reflect known relations among haematopoietic cell states.

For CARNIVAL, we focused on erythroid differentiation from the haematopoiesis dataset by comparing subnetworks inferred for megakaryocyte–erythroid progenitor (MEP) versus erythroid progenitor (ERP), multipotent progenitor (MPP), haematopoietic stem cell (HSC) and all contrasts. Subnetworks were encoded as edge-incidence vectors and projected into the principal component analysis space, with pairwise similarity matrices computed across regularization strengths (Supplementary Fig. 4b,c). Multi-sample CARNIVAL recovered signalling programmes that largely followed the expected developmental hierarchy. MEP–ERP, representing adjacent differentiation states, produced the most coherent and tightly clustered subnetworks, whereas MEP–MPP and MEP–HSC contrasts showed progressively greater divergence. The broad MEP–all comparison exhibited the weakest structural coherence, reflecting the aggregation of several haematopoietic lineages (Supplementary Note 2). These observations suggest that CARNIVAL captures regulatory relations that mirror developmental proximity and may, therefore, be useful for identifying signalling programmes associated with lineage commitment.

We next examined the multi-sample Steiner tree formulation using the same contrasts (Supplementary Fig. 5a–d). Like CARNIVAL, the MEP–ERP comparison yielded the most compact and coherent subnetworks, characterized by high connectivity among erythroid-associated regulators and low fitting errors. By contrast, MEP–MPP and MEP–HSC comparisons produced progressively larger and more diffuse subnetworks, consistent with increasing regulatory divergence from earlier progenitor states, whereas MEP–all generated the least specific structure (Supplementary Note 3). Unlike CARNIVAL, which emphasizes causal signalling consistency, Steiner tree highlighted the minimal network backbone required to connect differentiation-associated factors, providing a complementary view of the regulatory architecture underlying lineage transitions.

Taken together, the two approaches converged on the same biological interpretation despite relying on distinct optimization principles. Both recovered network structures are consistent with the known haematopoietic differentiation hierarchy, suggesting that multi-sample inference enhances the identification of robust lineage-associated regulatory programmes. CARNIVAL primarily resolved signalling relations and their consistency across contrasts, whereas Steiner tree emphasized the connectivity structure linking differentiation-associated regulators.

Strategies for improving robustness and interpretability

Our analyses identified two recurring challenges in constrained network inference: stochastic variability arising from optimization and information loss due to overly aggressive sparsification. Although these effects did not prevent CORNETO from recovering biologically meaningful structure, they can reduce reproducibility and obscure moderately informative features. We, therefore, investigated whether robustness and interpretability could be augmented through post-inference refinement strategies that operate on the inferred solutions rather than modifying the underlying optimization framework. To this end, we evaluated two complementary approaches: bootstrap aggregation to improve stability and selective re-expansion to recover informative biological structure.

Bootstrap aggregation improves stability and recall. We first investigated whether bootstrapped-based aggregation could reduce stochastic variability and improve the robustness of inferred networks. Using the sFBA framework with the MitoCore metabolic model and single-reaction knockout simulations, we repeatedly resampled intervention conditions and performed inference under both single-sample and multi-sample configurations. Reaction activation frequencies were then computed across bootstrap replicates, with reactions active in more >80% of runs classified as robust, and reactions recovered only through aggregation classified as expansion edges (Fig. 3a). Across all ensembles, both configurations achieved higher recall and network completeness than single-run baselines: in an eight-condition setting, bootstrapping recovered another 8–11 reactions per model (~8–11% expansion) while preserving more than 85% overlap with high-confidence baseline edges (Fig. 3b, top). These expansions complemented rather than replaced the original solutions, and reaction-level frequencies showed reduced stochastic variability, with ensemble-averaged solutions converging towards a stable subset

of shared reactions. Importantly, increasing the number of resampled conditions to 16 led to saturation, with little extra recovery of robust edges (Fig. 3b, bottom). At this point, single- and multi-sample configurations approached a fully constrained solution space, with the global overlap stabilizing at ~82% and ~89%, respectively. These findings indicate that bootstrap aggregation suppresses stochastic variability and consolidates reproducible network structure, with the strongest benefits observed in the multi-sample setting (Supplementary Fig. 6a,b).

Selective re-expansion recovers informative biological structure. We next examined a complementary challenge arising from aggressive sparsification. Although pruning substantially improves model compactness and interpretability, it can also remove moderately informative biological features that contribute to predictive performance. To determine whether such information could be recovered without abandoning sparsity, we evaluated a selective re-expansion strategy using the KPNN framework applied to the haematopoietic scRNA-seq dataset. Specifically, we reintroduced the top 100 most informative genes, ranked according to expression-based importance, that had been excluded from the pruned CORNETO directed acyclic graph. This targeted re-expansion increased the active edge set from 267 to 11,924, yet it remained substantially smaller than the original network. Importantly, predictive performance improved markedly: classification accuracy increased from 0.576 to 0.906, macro F_1 increased from 0.424 to 0.848, and both precision and recall exceeded 0.84 (Supplementary Table 2). These results suggest that the performance losses observed after pruning are not solely attributable to the reduced model size but also to the removal of moderately informative genes that collectively contribute to lineage discrimination. Selective re-expansion therefore provides a practical mechanism for restoring biologically relevant information while preserving much of the interpretability gained through network compression.

Better defined prior-network improves robustness. Beyond post-inference refinement, increasing prior-network completeness also improves constrained inference. Rather than artificially augmenting existing PKNs, we progressively removed edges from the Human-GEM reaction graph before multi-condition iMAT optimization, thereby simulating different levels of prior-network incompleteness. Decreasing the graph completeness removed held-out positive reaction-activity labels from the optimization graph, thereby reducing the recoverable evaluation signal (Fig. 3c). This loss of prior structure directly impaired precision–recall performance: at 20% graph completeness, all runs returned zero active reactions and produced no valid precision–recall points; at 40–60% completeness, the model recovered only a limited signal across λ values; whereas 80–100% completeness enabled higher precision and recall across a broader range of regularization strengths (Fig. 3d). These results show that prior-network completeness strongly determines the quality of the constrained metabolic inference and that regularization becomes useful only when sufficient prior structure remains available.

Increasing the number of jointly modelled conditions improves robustness. We next tested whether increasing the number of jointly optimized conditions stabilizes inference under repeated condition subsampling (Fig. 3e). Using iMAT, we randomly selected groups of 2, 4, 8 or 16 haematopoietic pseudobulk conditions and quantified active-reaction set stability using pairwise Jaccard similarity between active-reaction sets inferred from independent subsampling runs. Increasing the number of jointly inferred conditions generally increased the similarity between independently inferred solutions and reduced the sensitivity to the specific subset of conditions selected. Complementary analyses showed that increasing the number of conditions also stabilized the inferred network size and reduced the sensitivity to λ across repeated runs (Supplementary Note 4). Together, these results show that prior-network completeness and sample and/or condition availability jointly shape the robustness and reproducibility of constrained multi-condition network inference.

Together, these analyses demonstrate that pre- and post-inference refinement can substantially improve both robustness and biological interpretability without altering the underlying CORNETO optimization framework. Although demonstrated here in a subset of tools, all strategies reflect general principles that are broadly applicable across CORNETO inference tasks and data modalities. These results suggest that robustness can be improved not only through optimization design but also through a principled refinement of inferred solutions, providing a practical path towards more stable and biologically informative network reconstructions.

Discussion

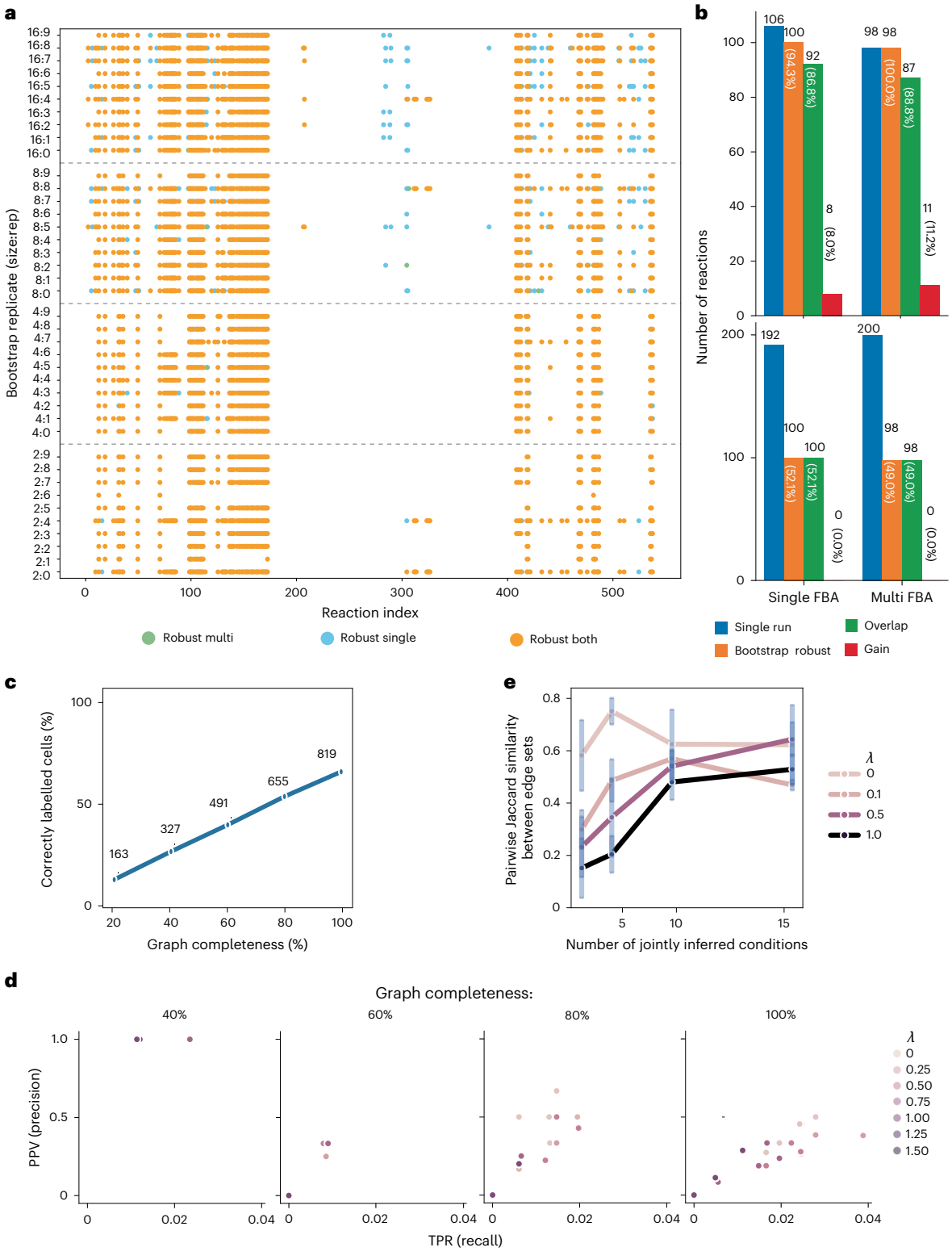
In this study, we reproduced and extended the original CORNETO framework, and we confirmed the reproducibility of its optimization pipelines across several inference paradigms. Consistent with the original work, all analyses followed the published benchmarking procedures, validation criteria and evaluation metrics. Our primary objective was not to establish new biological ground truths but to evaluate how multi-sample constrained inference affects robustness and stability across diverse data regimes.

Fig. 3 | Pre- and post-inference refinement can improve robustness. **a**, Reaction-level robustness across single-sample and multi-sample sFBA. Condition settings denote the number of knockout interventions jointly analysed in each experiment (2, 4, 8 or 16 knockout conditions), with 10 repetitions per setting and 20 bootstrap resamplings per repetition. Each point represents a reaction classified as robust in single-sample FBA only, multi-sample FBA only or both, using a robustness threshold of frequency >0.8. **b**, Comparison between reactions recovered in a representative single run and reactions retained after bootstrap aggregation Top: 8-condition setting. Bottom: 16-condition setting. In the 8-condition setting, bootstrap aggregation recovers additional robust reactions not present in the representative single run, whereas in the 16-condition setting the solution approaches saturation, with all bootstrap-robust reactions already represented in the single run and no additional gain. 'Single run' denotes reactions active in one non-aggregated run. 'Bootstrap robust' denotes reactions consistently recovered after bootstrap aggregation. 'Overlap' denotes reactions shared between the single run and the bootstrap robust set. 'Gain' denotes robust reactions recovered only after bootstrap aggregation. **c**, Effect of prior-network completeness on the availability of positive benchmark labels in the iMAT evaluation. The Human-GEM reaction graph was progressively downsampled from 100% to 20% completeness, and the percentage of held-out positive reaction-activity labels still represented in the optimization graph was quantified. Numbers indicate the number of reactions retained in the downsampled graph. **d**, Multi-condition iMAT precision–recall performance after prior-network downsampling. Points show PPV (precision) and TPR (recall) for different λ values across graph-completeness levels. At 20% completeness, all runs returned zero active reactions and, therefore, produced no valid precision–recall points. **e**, Condition-subsampling robustness of multi-condition iMAT in a multi-sample haematopoietic pseudobulk setting. Groups of 2, 4, 8 or 16 haematopoietic pseudobulk conditions were randomly sampled without replacement, with $n = 5$ independent computational subsampling runs performed for each condition set size and λ value. The unit of analysis was an independently sampled set of pseudobulk conditions; these repetitions represent computational resampling replicates rather than biological or technical experimental replicates. Active-reaction set stability was quantified by pairwise Jaccard similarity between inferred active-reaction sets. For each setting, the five independent runs yielded ten pairwise comparisons, and data are presented as mean pairwise Jaccard similarity ± s.d. across these ten comparisons. Higher similarity indicates reduced sensitivity to the specific subset of conditions selected. rep, repetition.

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Across iMAT, KPNN, Steiner tree and CARNIVAL, the principal benefit of multi-sample inference was improved robustness rather than uniformly improved predictive performance. Joint optimization reduced the sensitivity to stochastic variability and regularization by producing more stable and reproducible solutions when samples had a partially overlapping regulatory structure. However, these advantages depend strongly on data characteristics, including prior-network completeness and intersample heterogeneity, indicating that multi-sample coupling is most effective when sufficient shared structure exists across samples.

The haematopoietic and brain case studies illustrate these operating regimes. Multi-sample inference provided clear advantages for the dynamic haematopoietic system, whereas gains were more limited for the comparatively stable adult brain. These analyses were intended primarily to evaluate inference behaviour under distinct biological settings rather than to establish new regulatory mechanisms. More broadly, the lack of experimentally validated context-specific regulatory ground truths remains a principal limitation for network-inference benchmarking. Consequently, our conclusions focus on the robustness,



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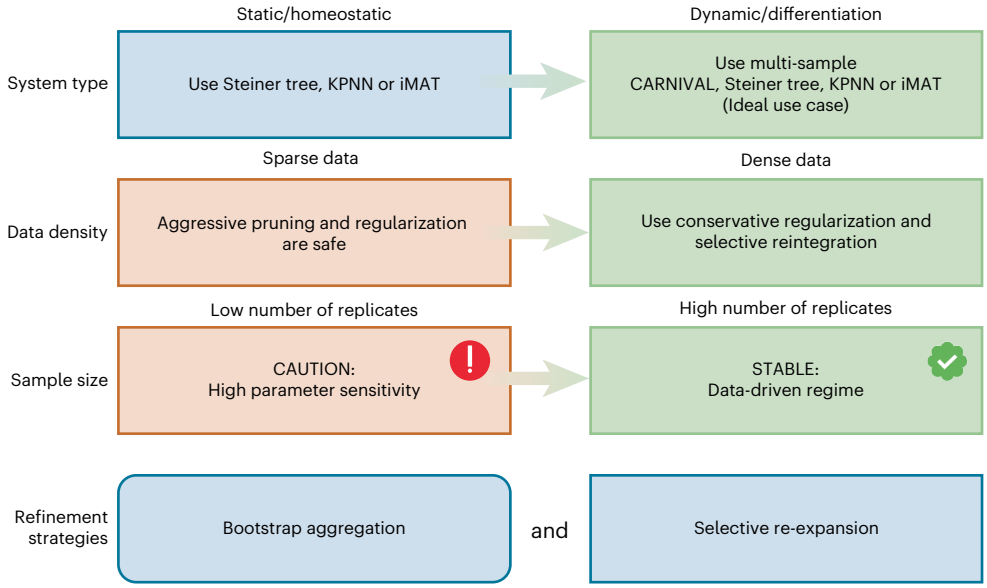


Fig. 4 | Case-study-informed factors influencing robustness in multi-sample constrained network inference. Decision matrix summarizing recommended inference tools, regularization strategies and refinements based on system dynamics, data density and sample availability. Dynamic, multi-sample settings favour joint inference, whereas static or sparse regimes require more conservative tool selection and regularization. Bootstrap aggregation and selective re-expansion improve robustness without altering the core optimization framework. The relations summarized here are conceptual principles derived from the computational perturbation analyses and the two biological case studies examined in this work. They are intended as practical hypotheses for CORNETO use rather than as systematically validated distinctions between dynamic and homeostatic biological systems.

reproducibility and identifiability of inferred solutions rather than realizing a definitive recovery of the causal regulatory interactions.

An important limitation of this study is its biological breadth. The contrasting dynamic and comparatively stable settings are represented by individual haematopoietic and cortical datasets and, therefore, constitute case studies rather than biological replicates of these broader system classes. Consequently, our observations regarding the influence of biological variability should be regarded as hypotheses about inference regimes rather than general properties of dynamic and homeostatic tissues. Establishing such generality will require systematic evaluation across several independent datasets, tissues, organisms and PKNs. Similarly, the biological analyses performed here assess consistency with established biological relations but do not constitute an experimental validation of the inferred causal regulatory mechanisms.

We show that robustness can be enhanced through both pre- and result-level refinement strategies. Bootstrap aggregation reduced stochastic variability and increased the recovery of reproducible network components, whereas selective re-expansion mitigated information loss from aggressive pruning. Additionally, a better prior network enhanced robustness. Together with the operating-regime framework summarized in Fig. 4, these results provide practical guidance for selecting regularization strategies and interpreting CORNETO outputs across different biological contexts.

This illustrates how data sparsity, prior completeness and sample availability jointly determine the stability of inference and provides actionable guidance for tuning CORNETO in specific biological contexts. Looking forwards, these principles motivate the development of adaptive regularization, data-driven sparsification and iterative refinement strategies embedded in agentic or human-in-the-loop workflows[37], positioning CORNETO as an evolving optimization backbone for scalable, knowledge-driven network inference[38–42].

Overall, our study highlights robustness as the primary advantage of multi-sample knowledge-constrained inference. By characterizing the conditions under which sample coupling improves stability and introducing practical strategies for enhancing robustness, we provide a framework for applying CORNETO more reliably across diverse biological settings[27].

Methods

Reproducing the original CORNETO framework

We reproduced the CORNETO framework using the official implementation available on GitHub (https://github.com/saezlab/corneto) and its companion repository containing all reproducibility scripts and benchmark configurations (https://github.com/saezlab/corneto-manuscript-experiments). All analyses were performed with CORNETO v.1.0.0, Python 3.12.10 and the Gurobi 12.0.2 optimization solver on a Linux high-performance computing environment. Solver precision and configuration parameters were kept identical to those in the reference study (absolute mixed-integer programming (MIP) gap of 1×10^{-6}). We adhered strictly to the parameterizations, objective formulations and graph specifications described in the original CORNETO article, fully replicating the canonical implementations for the four methods evaluated in this study: iMAT[18], KPNNs[25], Steiner tree[20] and CARNIVAL[19].

A comparative summary of the evaluated CORNETO paradigms, including inputs, outputs, optimization objectives, prior-network definitions and evaluation metrics, is provided in Supplementary Table 1.

sFBA. All analyses for the sFBA[16,17] task were performed using the MitoCore[43] v.1.0.1 metabolic model, a manually curated genome-scale reconstruction of human mitochondrial metabolism defining 324 metabolites and 553 reactions encompassing transport, exchange and mitochondrial biochemical processes. The model was obtained in the miom format used by CORNETO from the MIOM-GEMS repository (https://github.com/pablormier/miom-gems), ensuring full compatibility with the CORNETO framework for metabolic network optimization. In addition to the base model, a dataset of single-reaction knockout simulations (mitocore.csv) quantified the effect of each reaction on biomass production, including the reaction identifier, optimal biomass flux after knockout and the number of active reactions in the resulting minimal network. For the multi-condition sFBA

analyses, sets of interventions were systematically sampled and stored in experiments.csv, specifying the number of interventions per condition, replicate index and identifiers of the reactions removed in each experiment.

iMAT. We used the Yeast-GEM v.8.5.0 model, a genome-scale reconstruction of *Saccharomyces cerevisiae* metabolism obtained from the SysBioChalmers repository (https://github.com/SysBioChalmers/yeast-GEM). Gene–protein–reaction (GPR) associations and stoichiometric annotations were used to map transcriptomic perturbations from the Deleteome database (https://deleteome.holstegelab.nl/), which contains microarray-based expression profiles of yeast single-gene deletion mutants. Expression data were standardized, mapped to metabolic reactions using GPR rules and classified as significantly up- or downregulated relative to wild-type controls. The resulting matrices of differential expression values, significance scores and binary activation states were used as inputs for the multi-sample iMAT reconstruction and evaluation of CORNETO.

CARNIVAL. We used the PANACEA dataset (GSE186341), which provides transcription factor activity profiles across several cancer cell lines treated with targeted drugs. We selected the PONATINIB treatment and a subset of seven representative cell lines (H1793, LNCAP, KRJ1, HCC1143, EFO21, PANC1 and HF2597) to construct the multi-condition inference benchmark. Transcription factor activity scores were derived using the DoRothEA[44] resource and filtered for significance (adjusted $P \leq 0.05$). The CollecTRI PKN of signed protein–protein interactions (PPIs) was used as the structural backbone, and PONATINIB targets (BCR, ABL, VEGFR, FGFR and SRC) were modelled as inhibitory inputs. These data were used in the multi-condition implementation of CARNIVAL in CORNETO to reconstruct the context-specific signalling networks under perturbation.

Steiner tree simulation. We generated synthetic benchmark datasets to evaluate single- and multi-condition network optimization under various levels of overlap and complexity. Each dataset was based on a Barabási–Albert graph with 1,000 nodes and three new edges per added node, representing a sparse yet realistic network topology. Edges were assigned a uniform cost of 0.01, and node-specific prizes were sampled from a uniform distribution between 1 and 10 for a subset of terminal nodes. Three configurations were produced with increasing structural diversity: (1) 50 terminal nodes with 10 shared across all samples, (2) 20 terminal nodes with 5 shared and (3) 5 terminal nodes with no shared terminals. Each configuration included ten samples for cross-condition evaluation, with parameters stored in JSON configuration files (dataset_config.json) and resulting graph instances saved as CORNETO graph objects (graph_sample_i), providing a reproducible test bed for Steiner tree optimization.

Steiner tree for cytokines. As part of the original CORNETO benchmark, the Single-cell Perturbational (Cytokine Atlas Showcase) used the Cytokine Atlas dataset[45], comprising single-cell transcriptomic profiles of mouse immune cells stimulated with diverse cytokines and growth factors. Preprocessed pseudobulk differential expression data were obtained from the EMBL-EBI OwnCloud repository (cytokine_atlas_deseq.csv.gz). Genes were discretized as up- or downregulated (logFC > 0 and padj ≤ 0.05), and a mouse PKN was assembled from OmniPath10 interactions (mouse_allints.csv). For each cell type, the ten perturbations with the most differentially expressed genes were selected, and deregulated genes were assigned as terminal or prize nodes within the multi-condition PCST formulation in CORNETO. The optimization reconstructed compact subnetworks connecting perturbation-associated terminals through the shared prior interaction graph while balancing node prizes and edge costs across coupled conditions.

KPNN. We used the stimulation scRNA-seq dataset for T-cell receptors[34] from Fortelny and Bock[25]. The dataset, downloaded from those authors' repository (http://kpnn.computational-epigenetics.org/), includes expression matrices (TCR_Data.h5) from the 10x Genomics platform, binary annotations for T cell-receptor-stimulated and control cells (TCR_ClassLabels.csv), and a directed PKN connecting the receptor, signalling and transcriptional nodes (TCR_Edgelist.csv). Matched genes between the expression matrix and PKN were used to construct CORNETO-based KPNNs, where the PKN defines the directed acyclic graph, enabling interpretable deep learning.

Evaluation metrics and reference definitions

Our quantitative evaluation was performed using the same benchmarking procedures, validation criteria and reference definitions implemented in the original codebases and benchmark scripts provided by the authors of the respective methods and frameworks. Accordingly, our study did not redefine task-specific ground truths or introduce alternative evaluation criteria beyond those used in the original reference implementations.

In all analyses, PKN/KPNN was used as the constrained feasible search space for optimization, consistent with the original implementations, but was not interpreted as a universally biological ground-truth network.

Multi-condition iMAT. Predictive performance was evaluated using the same cross-validation strategy implemented in the original iMAT benchmark scripts, in which subsets of assigned reaction-activity states were masked before optimization and subsequently predicted from the remaining constraints. Predicted reaction-activity states were then compared against held-out labels derived from GPR-mapped transcriptomic constraints, enabling us to calculate the TPR (or recall), PPV (or precision), F_1 score, ROC-AUC and related classification metrics. Fivefold cross-validation was performed across ten random seeds for both single-sample ($\lambda = 0$) and multi-sample ($\lambda > 0$) configurations to evaluate performance across various coupling strengths. All computations were executed within the integrated R and Python workflow in CORNETO with an automatic export of intermediate matrices and optimization outputs for reproducibility.

Multi-condition CARNIVAL. This evaluation followed the same procedure implemented in the original benchmark code. A consensus reference solution was generated by averaging repeated baseline optimizations across several random seeds for a fixed regularization parameter. The fitting error was defined as the normalized discrepancy between inferred edge activation profiles and this consensus reference. Consequently, the reported proportion of edges from the PKN corresponds to the fraction of inferred active edges within the constrained signalling prior, consistent with the original implementation, rather than recovery against an independently validated biological regulatory ground truth.

Steiner tree. Similarly, this evaluation followed the original dataset-generation and benchmarking framework provided with the reference implementation. The interaction network defined the optimization backbone, and planted terminal prize structures defined the benchmark optimization targets. Performance was evaluated using reconstruction efficiency, terminal prize coverage, fitting objective and normalized edge usage, as implemented in the original benchmarking scripts. In the multi-sample formulation, subnetworks for several related conditions were jointly optimized over a shared interaction graph, and coupling penalties encouraged partial structural consistency between condition-specific solutions. Importantly, the effective size and topology of the interaction prior varied across biological conditions depending on the number and composition of the differentially expressed genes used to generate each reconstruction, as the global

human PPI network was subsetted before optimization following the original preprocessing workflow.

KPNN. This evaluation followed the original KPNN benchmarking framework. Predictive performance was assessed using classification accuracy, ROC-AUC, precision, recall and F_1 score across held-out validation data. The biological prior network constrained the neural network architecture by defining the permitted regulatory and signalling connections between input genes, hidden regulatory nodes and output classes. A KPNN-specific evaluation additionally assessed predictive reproducibility and the stability of inferred node-importance profiles across repeated training runs. Hidden-node-importance scores were quantified using integrated gradients and layer-wise relevance propagation, and ranked feature profiles were compared across pseudobulk conditions and independent retrainings to evaluate the consistency of the prioritized regulatory features.

sFBA. This evaluation followed the original sparse metabolic optimization framework implemented in the CORNETO benchmark scripts using the MitoCore metabolic reconstruction. Performance was quantified through the number of active reactions required to sustain near-optimal biomass production under simulated perturbations. In the multi-sample formulation, several perturbation conditions were jointly optimized, and sparsity penalties encouraged shared flux usage patterns across conditions. Robustness analyses further quantified reaction reproducibility across repeated bootstrap resampling runs using overlap-based stability metrics and reaction activation frequencies.

Accordingly, references throughout this Article to 'KPNN-consistent', 'PKN-consistent' or prior-constrained solutions should be interpreted as solutions recovered within the feasible search space defined by the corresponding prior biological network or constraint system following the original benchmark definitions and not as independently validated biological ground-truth interactions or universally true regulatory mechanisms.

Integrated framework analysis of the datasets

We expanded CORNETO benchmarking to two single-cell transcriptomic systems.

Haematopoietic differentiation. This dataset was derived from Ainciburu et al.[21]. This dataset profiles CD34$^+$ haematopoietic stem and progenitor cells from young (18–20 years) and elderly (>65 years) donors using the 10x Genomics Chromium platform, yielding over 75,000 high-quality single-cell profiles across 14 annotated haematopoietic subpopulations delineating the main differentiation trajectories of the human haematopoietic system. For the analyses performed in this study, we focused on six representative progenitor and lineage-associated populations spanning early and committed haematopoiesis: HSCs, MPPs, ERPs, MEPs, granulocyte–monocyte progenitors (GMPs) and common lymphoid progenitors (CLPs). We used this dataset to evaluate the performance, robustness and reusability of the multi-sample methods in CORNETO in capturing gene-expression programmes and network dynamics associated with complex haematopoietic differentiation processes. Throughout this Article, this dataset is referred to as the haematopoietic dataset.

Brain tissue. We used the Human Brain Atlas dataset from Siletti et al.[22]. This dataset includes over three million single nuclei from 46 cortical and subcortical brain regions, sequenced via single-nucleus RNA-seq and integrated to produce a unified taxonomy of neuronal and glial cell types. We extracted cell-type-resolved expression matrices and integrated them with the network-based modelling framework in CORNETO to construct region- and cell-type-specific regulatory subnetworks, thereby assessing the scalability and interpretability of CORNETO on large, heterogeneous single-cell datasets. We refer to this dataset as the human brain cortex dataset.

Together, these datasets span metabolic, signalling and transcriptomic contexts across several organisms and biological scales, providing a foundation for assessing the reusability, scalability and practical applicability of CORNETO across diverse network inference and optimization tasks.

Integrated framework analysis with iMAT

To examine the ability of CORNETO to generalize metabolic modelling beyond its original benchmark systems, we applied the multi-sample iMAT formulation to the haematopoietic and human brain cortex datasets. Both datasets contained raw single-cell transcriptomic profiles that were processed into pseudobulk expression matrices before optimization.

Data processing and pseudobulk construction. Single-cell data preprocessing was performed in R (v.4.3) using zellkonverter, SingleCellExperiment and Seurat. Count matrices were normalized with Seurat's LogNormalize function (scaling factor 10,000) and reduced to 2,000 highly variable genes selected via the vst method. Dimensionality reduction using a principal component analysis on these features enabled clustering and visualization, after which cell-type labels were transferred from the original metadata. In both datasets, normalized counts were aggregated by annotated cell type to form pseudobulks. For the haematopoietic dataset, pseudobulks captured distinct differentiation stages, including HSC, GMP, monocyte–dendritic progenitor (MDP), ERP, CLP and Pre–B cell (PreB) populations. In the human brain cortex dataset, pseudobulks represented the main cellular classes (fibroblasts, endothelial cells, astrocytes, oligodendrocytes, neurons, pericytes, leukocytes, central nervous system macrophages and oligodendrocyte precursor cells) averaged across brain regions. All gene identifiers were mapped to HGNC symbols using org.Hs.eg.db, and only genes present in the Human-GEM metabolic model were retained.

Differential expression and gene activity mapping. For each cell type, we performed a differential expression analysis against all other cells using the Wilcoxon rank-sum test (FindAllMarkers in Seurat) with a detection threshold of 0.25, $\log_2$ fold-change ≥0.25 and adjusted $P \leq 0.01$ (Benjamini–Hochberg correction). The resulting matrices of $\log_2$ fold-change, adjusted P values and mean expression were exported for downstream iMAT reconstruction.

Metabolic models in the PKN. All metabolic models were based on Human-GEM v.1.9.0 (https://github.com/SysBioChalmers/Human-GEM), comprising 13,417 reactions, 5,686 metabolites and 3,741 genes.

Method. GPR rules were used to propagate transcriptomic states to reactions. Genes were classified as active, inactive or uncertain using adaptive percentile thresholds: adjusted $P \leq 0.05$ and $\log_2$ fold-change above the 70th percentile indicated activation, whereas genes below the 30th percentile were marked inactive. Boolean logic was applied to GPR associations to derive condition-specific reaction-activity scores, generating discrete and continuous formulations for iMAT input.

Multi-sample optimization. The multi-sample iMAT implementation jointly optimized reaction states across pseudobulks while penalizing deviations between samples through a coupling term weighted by λ. Each pseudobulk was treated as a distinct sample within a unified mixed-integer linear program solved with Gurobi (v.12.0.2). The solver tolerance ε was set to 10^{-3}, with mixed-integer programming gaps between 0.05 and 0.10. Regularization strengths spanned $\lambda \in \{0, 0.01, 0.05, 0.1, 0.25, 0.5, 0.75, 1, 1.5\}$, and time limits ranged from 120 s to 5,000 s depending on λ and dataset size.

Integrated framework analysis with the multi-sample Steiner tree

To test the reusability of CORNETO for topological network reconstruction tasks, we implemented a multi-sample variant of the PCST model. This formulation evaluates how well CORNETO infers optimal subnetworks from differential expression data while maintaining shared structural constraints across related samples. The analyses were carried out on the same two human datasets already described (the haematopoietic differentiation and brain cortex scRNA-seq collections), which were processed identically.

Differential expression and graph preparation. For each dataset, haematopoietic and human brain cortex, the differential expression between each cell type and all others was calculated using the Wilcoxon rank-sum test. Genes with adjusted $P \leq 0.05$ and $|\log_2 \text{fold-change}| \geq 0.5$ were retained as significantly modulated. For every cell population, the resulting tables of $\log_2$ fold-change and significance values provided the basis for assigning node-level prizes in the PCST model.

PPIs in PKN. For the PKN, we retrieved all human PPIs from OmniPath, providing a comprehensive network of 282,505 curated interactions to be used in subsequent analyses. Graphs were built in NetworkX, containing 20,109 nodes and 268,165 edges overall. For each cell population, subnetworks were restricted to differentially expressed genes mapped to the PPIs.

Single-sample optimization. The single-sample baseline used the exact_steiner_tree() solver in CORNETO to find the minimal-cost subgraph that maximized the total prize collected. For each cell population, the differentially expressed genes were ranked by their absolute $\log_2$ fold-change values, and the top 5, 20 and 50 genes were selected as terminals for independent runs. Each node received a prize proportional to its absolute $\log_2$ fold-change, whereas edges were uniformly penalized with a cost of 0.01 to discourage excessive connectivity. Optimization was performed with Gurobi (v12.0.2) under a 600-s time limit, integer tolerance of 10^{-6} and a fixed random seed (42). Each λ value in the set {0, 0.1, 0.15, 0.2, 0.25, 0.3, 0.35, 0.4, 0.5, 1, 2, 3, 5} produced a distinct subnetwork, which was stored alongside summary statistics, including runtime, node and edge counts, and the collected prize fraction for downstream comparisons.

Multi-sample optimization. The multi-sample extension used the same parameters but introduced a coupling term to jointly optimize subnetworks across samples sharing a common PPI structure. Genes not identified as differentially expressed in all samples were excluded to ensure consistent node representation across the dataset. Implemented via the create_exact_multi_steiner_tree() routine, the optimization balanced dataset-specific variation and cross-sample similarity through the λ regularization coefficient. To assess the ability of CORNETO to regenerate biologically meaningful networks, all differentially expressed genes identified for each population were used as terminals to generate a multi-sample Steiner tree without restricting the terminal count. The resulting networks were examined for prize coverage, edge overlap across samples and the presence of known regulators.

Computational set-up. All experiments were executed under identical solver settings to ensure comparability. Uniform edge costs of 0.01 were used, and candidate terminal nodes were defined from the differential-expression preprocessing described above. Analyses were conducted using Python 3.11, NetworkX and CORNETO v.1.0 with Gurobi 12.0.2 on the IBEX High-Performance Computing Cluster at King Abdullah University of Science and Technology (KAUST). Output directories, intermediate MILP models and graph artefacts were automatically organized and archived for reproducibility.

Integrated framework analysis with KPNN

Data processing and input preparation. Both datasets, haematopoietic and human brain cortex, were preprocessed exactly as described for the iMAT and Steiner tree analyses, but data were used at the single-cell level (no pseudobulks). Genes were retained if they were present in the reference signalling and regulatory networks used to define the KPNN topology.

Model architecture and biological prior construction. The KPNN architecture was defined using a prior biological network that connects input genes to internal regulatory nodes (representing transcription factors, receptors and signalling intermediates) and to output nodes corresponding to cell-type or condition labels.

In the KPNN analysis, the goal was not to impose an identical prior graph across biologically distinct tissues but to apply the same prior-constrained modelling principle in each dataset. KPNNs require that input genes be embedded into a directed biological network connecting genes, regulatory/signalling intermediates and output labels. Therefore, the effective KPNN prior necessarily depends on both the biological system under study and the overlap between measured genes and curated regulatory resources.

For the haematopoietic dataset, we constructed a haematopoiesis-oriented regulatory prior by integrating signalling interactions from SIGNOR with transcriptional regulatory relations from Harmonizome TFBS/ChIP-seq-derived resources. This choice was motivated by the strong role of transcription factor programmes and signalling cascades in haematopoietic lineage commitment. Genes that were present in the expression matrix but could not be connected to the regulatory or signalling layers were excluded because they could not be represented in the directed KPNN graph. After expression-network alignment and pruning of disconnected components, the final haematopoietic prior contained 9,791 input genes connected to at least one regulatory or signalling node.

For the human brain cortex dataset, we constructed a second curated human regulatory/signalling prior using Reactome, TRRUST and KEGG. Reactome provides manually curated and peer-reviewed pathway knowledge, including signalling, metabolism, transcriptional regulation, apoptosis and synaptic processes, making it particularly suitable for brain tissue, as pathway-level and synaptic signalling annotations are important. TRRUST provides literature-curated transcription factor–target interactions and was, therefore, used to define interpretable transcriptional regulatory edges. Self-loops, redundant edges and unsupported interactions were removed, yielding a sparse directed graph of approximately 2,800 nodes and 10,000 edges.

Thus, the two KPNN priors differed in their database composition because the datasets represented distinct biological systems and had different gene-coverage constraints. However, the modelling procedure was kept consistent across datasets: in both cases, we used curated human regulatory/signalling knowledge, retained only supported directed interactions, removed disconnected or unsupported genes, mapped expressed genes to prior-network input nodes, and trained separate KPNN models with the same overall architecture and hyperparameterization. This design preserves interpretability while avoiding an artificial one-size-fits-all prior that would either discard many informative dataset-specific genes or introduce poorly supported edges.

Hidden nodes represented regulatory and signalling intermediates constrained by the biological prior network. To facilitate interpretation of the learned representations, node-importance scores were estimated using integrated gradients and layer-wise relevance propagation.

Training procedure and regularization. KPNN models were implemented in TensorFlow 2.15 and trained using a sigmoid-activated multilayer architecture corresponding to the prior network structure.

Input nodes represented normalized gene-expression features, hidden nodes corresponded to regulatory proteins or transcription factors, and output nodes encoded cell-type probabilities. We used the categorical cross-entropy loss for each classification task (cell type versus all others) and trained the network using the Adam optimizer with a learning rate of 1×10^{-3} and a batch size of 64. Dropout regularization ($P = 0.2$) was applied to all hidden layers to prevent overfitting, and L1 weight penalties were used to encourage sparsity in network activation patterns. Early stopping was triggered if the validation loss failed to improve for 15 consecutive epochs, with a maximum of 300 epochs per model. For each dataset, fivefold cross-validation was performed using 80% for training and 20% for validation, repeated over five random seeds to ensure reproducibility. Training stability and convergence were monitored by tracking validation loss, accuracy and balanced F_1 score across epochs. Final weights were averaged across folds to generate consensus parameter matrices. Each trained KPNN was serialized to disc, preserving both learned weights and structural metadata for interpretability analysis.

Computational set-up. All training and evaluation tasks were conducted in Python 3.11 with TensorFlow, NetworkX and NumPy. Each experiment used fixed random seeds and identical hyperparameters to ensure reproducibility. Results, including trained weights, node-importance scores and performance metrics, were automatically logged by the benchmarking interface of CORNETO for subsequent aggregation and comparative analysis alongside the iMAT and PCST tasks.

Integrated framework analysis with CARNIVAL

Data processing and input preparation. Both datasets, haematopoietic and human brain cortex, were preprocessed exactly as described for the iMAT and Steiner tree analyses. For each pseudobulk, differential expression was computed against all remaining cell types using PyDESeq with donor as a blocking factor, yielding $\log_2$ fold changes and adjusted P values for the corresponding one-versus-rest comparison. Upstream receptor nodes were selected from significantly regulated ligand–receptor genes, and transcription factor activities were computed using the univariate linear model in decoupler-py[46] on the CollecTRI gene regulatory network retrieved from OmniPath. Each sample contributed one input–output contrast, in which upregulated receptors were designated as activating inputs and transcription factors with positive or negative differential activity were assigned corresponding output signs. The resulting dataset comprised four donor-specific contrasts and was serialized for reproducible execution. Two representative samples were retained in the main runtime and λ-sensitivity analyses to ensure comparability with other reusability benchmarks. For the human brain cortex dataset, pseudobulk contrasts were constructed between oligodendrocyte progenitor cells and mature oligodendrocytes, and other pseudobulks were generated for other principal cell classes to assess computational scaling up to nine biological contexts.

PKN. All causal network reconstructions were based on the CollecTRI prior network, imported as a directed graph using the parser within CORNETO. The network consisted of approximately 7,500 nodes and 90,000 directed edges encompassing curated signalling and transcriptional relations from OmniPath, Reactome[47] and TRRUST. Before optimization, the network was pruned to include only nodes that matched the observed input and output genes in each dataset, thus reducing complexity and focusing inference on relevant subspaces of the graph.

Optimization set-up for single samples. Independent single-condition optimizations were performed with the milp_carnival solver in CORNETO using Gurobi 12.0.2 as a back end. Each integer linear programming sought a minimal-cost subnetwork consistent with the observed transcriptional responses. Solver parameters were set to a maximum runtime of 120 s, a heuristic warm-start period of 120 s and an integrality focus of 1. The regularization parameter β, which controls the trade-off between sparsity and fit, was varied between 0 and 1 to scan different penalty strengths. Ten replicates were executed with different random seeds, and for each, the resulting solution was evaluated by measuring the normalized output error and the number of active edges, providing an estimate of accuracy and network compactness.

Optimization set-up for single samples. Multi-sample optimization was implemented through the flow-based CARNIVAL engine in CORNETO, which builds a coupled integer linear programming that simultaneously infers subnetworks for several related samples. In this multi-condition setting, CORNETO introduced a coupling regularization term λ that penalizes disagreement in edge activation patterns between subnetworks representing related conditions. The optimization minimized the sum of per-sample costs and the intersample discrepancy term scaled by λ. Regularization strengths between 0 and 1 were explored to test the effect of increasing coupling. Solver configurations were identical to the single-sample set-up, using 120 s as the time limit and four computational threads per run. Each experiment was repeated ten times with different random seeds to evaluate robustness. CORNETO internally expanded the causal network to align all sample-specific layers and constructed the coupled optimization graph using its create_flow_carnival_v3 formulation. The Gurobi solver returned both edge activation variables and vertex activities across all samples in a single optimization pass.

Computational set-up. All analyses were conducted in Python 3.11 using CORNETO v.1.0 and Gurobi 12.0.2 on the KAUST IBEX High-Performance Computing Cluster. Reproducibility was ensured through fixed random seeds and configuration tracking. Intermediate artefacts, including pruned networks, serialized integer linear programming models and per-sample score matrices, were automatically logged and stored using the integrated experiment management framework in CORNETO.

Extended reusability analysis, graph completeness and sample scaling in multi-sample iMAT

To evaluate the robustness and scalability of the multi-sample iMAT formulation of CORNETO, all extended analyses were conducted using the haematopoietic scRNA-seq dataset from Ainciburu et al.[21]. This dataset captures differentiation across the haematopoietic lineages, providing six biologically distinct yet developmentally related pseudobulk conditions: HSC, GMP, MDP, ERP, CLP and PreB. The same preprocessed expression matrices, gene–reaction mappings and Human-GEM v.1.9.0 metabolic model used in the main iMAT reusability tests were reused here to ensure methodological continuity.

Graph-completeness perturbation. To assess the sensitivity of multi-sample metabolic inference to incomplete prior knowledge, the full Human-GEM network was systematically downsampled to simulate different degrees of structural completeness. Five completeness levels were generated by retaining 100%, 80%, 60%, 40% or 20% of the original reaction–metabolite edges through uniform random sampling while preserving the connectivity of all reaction nodes. We verified that each resulting subgraph had a single connected component encompassing all metabolic pathways relevant to the six pseudobulk samples.

For each graph-completeness fraction, the same multi-sample iMAT optimization pipeline was executed under the original configuration. The optimization problem was formulated as a mixed-integer linear program and solved using Gurobi v12.0.2 with a numerical tolerance $\varepsilon = 10^{-3}$, MIP gap of 0.05, and time limits between 120 s and 5,000 s, depending on model size. Regularization strengths were explored over a grid of $\lambda = \{0, 0.01, 0.05, 0.1, 0.25, 0.5, 0.75, 1, 1.5\}$, allowing us

to analyse how intersample coupling interacted with variations in network completeness.

After each run, model outputs were evaluated through fivefold cross-validation, in which 20% of gene activity entries were masked and re-estimated during the optimization. Accuracy, precision, recall and F_1 score were computed on the held-out entries. Additionally, the fraction of reactions retained relative to the reference full-network solution obtained from the complete Human-GEM model was quantified to measure structural degradation. These metrics allowed a direct assessment of how reduced connectivity limits feasible flux distributions and impacts model predictive power.

Sample size scaling. To examine how the number of input samples influences optimization stability and computational efficiency, the same haematopoietic dataset was used to construct incrementally larger multi-sample configurations. Starting from a single pseudobulk (HSC), the number of conditions included was progressively increased to two, three, four, five and finally six pseudobulks (HSC, GMP, MDP, ERP, CLP and PreB). Each configuration reused identical data matrices, thresholds and solver parameters as the base case, ensuring that changes in performance reflected only sample scaling effects.

For each subset size, the model was optimized across the full λ grid. The number of decision variables, solver runtime and convergence statistics were recorded to quantify computational scaling. Predictive accuracy, intersample coherence (mean Jaccard similarity of active-reaction sets) and solution sparsity were computed as before. These analyses quantified both algorithmic scaling and the stability of multi-sample coupling under increasing model dimensionality.

Sample subsampling and stability analysis. To systematically evaluate how condition availability influences the stability of multi-sample inference, we performed a progressive subsampling analysis using the haematopoietic pseudobulk dataset. This analysis was designed to quantify how the number of jointly optimized biological conditions affects reconstruction robustness under various regularization strengths.

From the full set of available haematopoietic pseudobulk conditions, subsets of 2, 4, 8 or 16 conditions were randomly sampled without replacement. For each condition set size and each regularization value, five independent computational subsampling runs were generated using distinct random seeds.

Each sampled subset was optimized using the same multi-sample iMAT configuration, Human-GEM v.1.9.0 metabolic model, preprocessing pipeline and solver parameters described above. Optimization was performed across the regularization values $\lambda \in \{0, 0.1, 0.5, 1.0\}$ using Gurobi v.12.0.2 with identical convergence tolerances and runtime limits.

To quantify reconstruction stability, binary active-reaction sets inferred from the five independent subsampling runs were compared using pairwise Jaccard similarity coefficients. The five runs yielded ten pairwise comparisons for each condition set size and λ configuration. Stability was summarized as the mean pairwise Jaccard similarity, and variability was quantified as the standard deviation across the ten pairwise comparisons. Network size and other optimization outputs were also recorded to evaluate how the number of conditions influenced structural reproducibility and inference behaviour.

This analysis enabled a direct assessment of whether increasing the number of jointly optimized conditions stabilizes network reconstruction and reduces sensitivity to regularization. All outputs, including reaction-level activity matrices, pairwise similarity scores and aggregated summary statistics, were stored for a downstream visualization and robustness analysis.

Metric aggregation and output processing. For both robustness dimensions—graph completeness and sample scaling—results were aggregated across λ values and random seeds. Average precision, recall and F_1 scores were computed for each configuration and plotted as functions of completeness and sample count. Structural stability was assessed by comparing reconstructed reaction sets between runs using the pairwise Jaccard similarity. Computational performance metrics, including solver time, variable count and feasible solution fraction, were also summarized.

All analyses were implemented in Python 3.11 using the CORNETO v.1.0 framework, NumPy, pandas and Matplotlib and executed on the IBEX High-Performance Computing Cluster at KAUST. Random seeds and solver configurations were fixed across all experiments to ensure strict reproducibility. Outputs—including filtered graph snapshots, summary CSVs and serialized mixed-integer linear program models—were stored in structured directories for post-analysis and visualization.

Computational scalability and resource assessment

To assess computational scalability and practical resource requirements across CORNETO paradigms, we monitored optimization runtime, solver convergence behaviour, memory usage and scaling with the number of samples and prior-network size for representative iMAT, CARNIVAL, Steiner tree and KPNN benchmark configurations. All mixed-integer optimization problems were solved using Gurobi v12.0.x on the KAUST IBEX High-Performance Computing Cluster unless otherwise specified.

For multi-sample iMAT, representative optimization instances contained approximately 50,000 decision variables after the pre-solve step and converged within ~25–45 s per optimization. Complete benchmark configurations comprising nine regularization settings ($\lambda \in \{0, 0.01, 0.05, 0.1, 0.25, 0.5, 0.75, 1, 1.5\}$), fivefold cross-validation and ten random seeds required approximately 3–6 h of cumulative compute time. The computational cost increased primarily with the number of jointly optimized samples and the size of the constrained metabolic prior network.

For multi-sample CARNIVAL, computational benchmarking was performed using the haematopoiesis differentiation dataset (MEP versus all contrasts and four donor samples) with CollecTRI as the signalling prior network. Runtime and memory usage were evaluated across single-condition and multi-condition formulations while we varied the number of jointly inferred samples ($n \in \{1, 2, 3, 4\}$), prior-network size through edge subsampling (fractions {0.5, 0.75, 1.0}), sparsity regularization ($\lambda \in \{0.1, 0.25, 0.5, 0.9\}$) and three random seeds. All problems were solved using four Gurobi threads with a 300-s solver time limit and 60-s NoRelHeurTime. Across successful runs, multi-condition CARNIVAL completed in a median runtime of ~90 s for four jointly inferred samples, whereas equivalent repeated single-condition analyses required ~315 s cumulatively. The peak resident memory usage remained comparable between formulations (~1.2–1.3 GB median resident set size at $n = 4$), although single-condition runs exhibited greater variance and occasional peaks approaching ~3.1 GB. The runtime showed a minimal dependence on λ but increased with prior-network size and number of jointly optimized conditions. Edge subsampling to 50% frequently disrupted the feasible signalling paths, leading to infeasible optimization instances in the multi-condition formulation.

For multi-sample Steiner tree inference, scalability was evaluated by varying both the candidate prior-network size and the number of jointly optimized samples. Representative benchmark instances ranged from ~50 to ~500 nodes, with approximately 20 terminal nodes per sample and sample counts ranging from 4 to 50 coupled conditions. Small prior networks (~50 nodes and ~70 edges) remained computationally tractable, requiring ~10–60 s for 4–20 jointly inferred samples and ~230–600 s for 50 samples. Intermediate networks (~100 nodes and ~180–200 edges) required substantially longer runtimes and frequent solver-limit behaviour, particularly for ≥10–20 samples. Larger prior networks (~250 nodes and >700 edges) consistently approached or reached the imposed 600-s solver limit, even with four coupled

samples, whereas ~500-node networks uniformly exceeded the time limit across the configurations tested. Computational complexity increased nonlinearly with the number of jointly optimized samples, terminal-set size and prior-network density, reflecting the combinatorial growth of the coupled mixed-integer optimization problem. Memory usage also increased substantially with network size and occasionally reached the gigabyte range in the largest instances. For very large formulations, CVXPY canonicalization itself became a computational bottleneck before the Gurobi optimization.

For the KPNN analyses, the wall-clock runtime was recorded separately during neural network training and CORNETO-based pruning. The initial KPNN training stage used a neural network architecture based on a directed acyclic graph with batch normalization, dropout, sigmoid hidden and output activations, and a softmax classification head trained using fivefold cross-validation for up to 200 epochs with early stopping and a batch size of 64. Initial model training required approximately 176 min. CORNETO-based pruning was subsequently formulated as an acyclic-flow mixed-integer linear program solved primarily using SCIP with sink-coverage constraints and edge-minimization objectives. The pruning stage required approximately 243 min and represented the dominant computational bottleneck. KPNN experiments were performed on a workstation equipped with an NVIDIA RTX 8000 GPU (48 GB memory) and dual Intel Xeon Gold 6230R CPUs (52 cores in total). Although neural network training benefited from GPU acceleration, pruning complexity was dominated by discrete mixed-integer optimization.

Result-level enhancement due to bootstrapped sFBA expansion. To assess and improve the robustness of flux-based predictions made by CORNETO, we implemented a bootstrapped sFBA framework designed to quantify the stability of inferred reactions and identify reproducible yet subthreshold interactions across repeated resamplings. All analyses were conducted on the haematopoietic metabolic model derived from the MitoCore v.1.0.1 network, which was used as the flux optimization backbone. The experiments were performed on the IBEX High-Performance Computing Cluster using Python 3.11, CORNETO v.1.0 and the Gurobi 12.0.2 solver.

Model set-up. The metabolic graph was constructed from the MitoCore v.1.0.1 model containing 647 reactions and 446 metabolites and encoded as a bipartite graph within CORNETO. The biomass export reaction (EX_biomass_e) was used as the objective function, and the associated flux variable served as the quantitative measure of system viability. All flux bounds were initialized using the default lower and upper limits in the model. Optimization was conducted under two configurations: a single-sample sFBA, in which each perturbation was solved independently, and a multi-sample sFBA, where all perturbations were jointly optimized with a shared sparsity constraint to enforce consistency across samples. In both cases, sparsity was introduced through binary indicator variables to enforce near-minimal flux activation per reaction. The solver tolerance was fixed at 10^{-6}, and all optimization problems were solved using mixed-integer programming with integrality focus enabled.

Intervention definition. Single-reaction knockouts were simulated by constraining fluxes to zero for each reaction individually. For each knockout, a baseline optimization was performed to compute the maximum achievable biomass flux followed by a sparse optimization that enforcing a minimum proportion of this flux (90%) to identify the minimal number of active reactions compatible with near-optimal growth. Reactions whose removal reduced the feasible objective by more than 0.1% but did not completely abolish the flux were retained as informative interventions. This filtering produced the subset of approximately 200 perturbations used for the bootstrap analyses, referred to as the intervention set.

Bootstrap sampling and execution. To quantify stability, a bootstrap resampling procedure was applied in which subsets of the intervention set were repeatedly sampled with replacement. For each bootstrap replicate, sFBA was executed under both single- and multi-sample configurations using the same solver parameters and sparsity regularization ($\alpha = 1.0$). A total of 20 bootstrap replicates ($B = 20$) were performed for each experimental configuration. The number of simultaneous intervention conditions per bootstrap iteration was varied among 2, 4, 8 and 16, enabling an evaluation of how the diversity of perturbations influenced robustness and coverage.

For each replicate, binary reaction-activity matrices were generated by thresholding the optimized flux magnitudes at $|\text{flux}| \geq 10^{-6}$. Reaction-level activation frequencies were computed across bootstrap replicates, producing 'robustness frequency' profiles. Reactions active in at least 80% of bootstrap replicates (frequency > 0.8) were classified as robust, whereas those emerging only after aggregation (not active in individual runs but appearing after frequency averaging) were designated as expansion edges.

Statistical quantification and robustness metrics. To quantify reproducibility, pairwise overlaps between robust reaction sets across bootstrap replicates were computed using the Jaccard similarity, and the mean overlap within each subset size was used to evaluate within-group consistency. The global overlap, defined as the intersection over the union of all robust reaction sets across all bootstrap runs, measured the degree of convergence between independent ensembles. Across all configurations, the global overlap stabilized at approximately 82% for single-sample sFBA and 89% for multi-sample sFBA, indicating higher reproducibility under the multi-sample coupling scheme.

Reaction-level statistics were also aggregated across subset sizes. For each reaction, the median, mean and standard deviation of its activation frequency across bootstrap replicates were computed separately for single- and multi-sample settings. Other stability indices, such as the robust count (number of replicates exceeding the 0.8 threshold) and robust score (product of mean frequency and normalized replicate count), were used to identify highly stable reactions. These metrics were summarized across the ten independent repetitions per subset size and exported as per-reaction and per-size summary tables.

Computational scaling and convergence. Each optimization run was executed with a 400-s time limit and default Gurobi heuristics (MIPGap = 0.01). Warm-start options were used when available to accelerate convergence between iterations. The solver runtime, number of binary variables and convergence status were recorded for each configuration and used to characterize the computational scalability.

The bootstrap frequency matrices, reaction-level stability summaries and graphical overlays depicting robust reactions across bootstrap replicates were generated using NumPy, pandas and Matplotlib. Statistical analyses were performed using Python-based data pipelines, and all outputs—including bootstrap result tables, robustness profiles and visualization PDFs—were archived for the reproducibility and downstream analysis.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The haematopoiesis dataset is available via the Gene Expression Omnibus under accession GSE180298 and was originally reported by Ainciburu et al.[21]. The brain single-cell dataset is available via the CZ CELLxGENE Discover portal (https://cellxgene.cziscience.com/collections/283d65eb-dd53-496d-adb7-7570c7caa443) and corresponds to the dataset reported by Siletti et al.[22]. All other datasets used for the reproducibility analyses are those described and provided with the original CORNETO framework[14]. Source data are provided with this paper.

Code availability

The code supporting this study is available via GitHub at https://github.com/TranslationalBioinformaticsUnit/Reusability-report-CORNETO-framework-for-knowledge-driven-multi-sample-network-inference. The specific version of the code associated with this study is available via Zenodo at https://doi.org/10.5281/zenodo.22229425 (ref. 48).

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Author contributions

X.M.-d.-M. designed the study, drafted the paper, performed the reproducibility analyses, conducted the mathematical and computational robustness analyses, and generated figures. X.M.-d.-M., S.A.K., I.C. and R.L. performed the reusability analyses and contributed to the evaluation of CORNETO across tools and datasets. X.M.-d.-M. and S.A.K. performed the enhancer analyses and contributed to methodological refinements. A.M., V.L. and I.C. contributed critical feedback on the text and figures. J.T. and D.G.-C. supervised the project and contributed to the editing and final revisions of the paper. All authors reviewed and approved the final Article.

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Competing interests

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